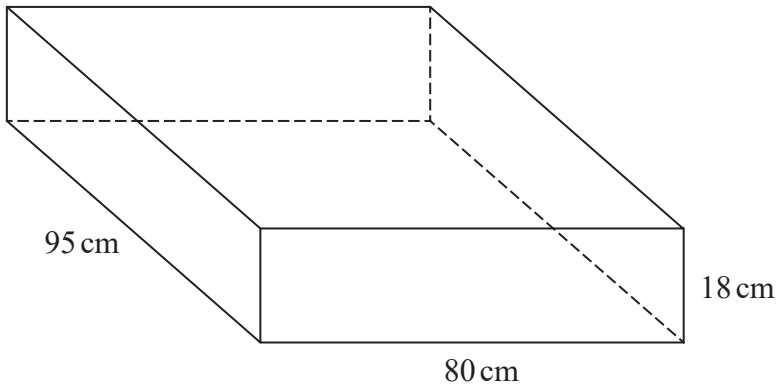
19 A sofa has 6 identical cushions.
Each cushion is a cuboid 18 cm by 80 cm by 95 cm.



The cushions are covered with a protective spray.
The protective spray is in cans.

The label on each can has this information.

Spray in this can covers 4 m^2

(a) Work out how many cans are needed to cover the 6 cushions with protective spray.

Convert first: 18cm=0.18m; 80cm=0.80m; 95cm=0.95m.

1 mark
Consistent square metres

Face areas: $0.18 \times 0.80 = 0.144\ m^2$;

$0.18 \times 0.95 = 0.171\ m^2$; $0.80 \times 0.95 = 0.760\ m^2$.

1 mark
Three face types

One cushion = $2(0.144 + 0.171 + 0.760) = 2.15\ m^2$.

1 mark
Double all face types

Six cushions = $6 \times 2.15 = 12.9\ m^2$.

1 mark
Six areas ÷ coverage

Cans = $12.9 \div 4 = 3.225$; buy4 cans.

1 mark
4 cans; round up

(5)

The information on each label is inaccurate.
The spray in each can covers 10% more than $4\,m^2$.

(b) How will this affect the number of cans needed for the 6 cushions?
You must show how you get your answer.

New coverage = 4 × 1.10 = 4.4 m^2 per can.

1 mark
10% more coverage

12.9 ÷ 4.4 = 2.9318...; buy3 cans.

The number needed reduces from4 cans to3 cans.

1 mark
Reduces to3

(2)

(Total for Question 19 is 7 marks)